AP BIOLOGY TOPIC 3.2

Environmental Impacts on Enzyme Function, Kinetics & Inhibition

UNIT 3: CELLULAR ENERGETICS (12–16% EXAM WEIGHT)

ENE-1.F | ENE-1.G | DENATURATION & V_{max} / K_m

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 3.2 (ENVIRONMENTAL IMPACTS ON ENZYME FUNCTION, KINETICS & INHIBITION)

CED Core Vocabulary

Structure → Function Relationship — Enzyme catalytic activity depends strictly on its native 3D folding (secondary 2° , tertiary 3° , and quaternary 4° structure) , which is stabilized by weak...

Hydrophobic interactions — (nonpolar core) + covalent disulfide bridges (–S–S–)

Reversible vs. Irreversible Denaturation — Reversible (Renaturation): Mild environmental shift; once optimal conditions are restored, the polypeptide spontaneously refolds into its active shape.

Temperature Effects (Asymmetric Curve) — Below Optimum (↑ Temp → ↑ Rate): Higher thermal kinetic energy increases molecular velocity and frequency of successful enzyme-substrate collisions .

Ionic interactions / salt bridges — (charged acidic & basic R-groups)

Denaturation — Loss of native 3D protein conformation caused by extreme temperature, pH, salinity (ion concentration), or chemical denaturants → active site is distorted →...

Irreversible — Extreme heat or pH causes permanent aggregation/precipitation (e.g., boiling an egg white albumin).

Above Optimum (Rapid Drop) — Excessive thermal vibration breaks weak H-bonds and hydrophobic interactions → thermal denaturation → steep plunge in activity.

§2. STRUCTURE → FUNCTION & ENZYME DENATURATION

3D Folding Integrity

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Structure → Function & Enzyme Denaturation ➔ **Structure → Function Relationship** ➔ **Denaturation** ➔ **Reversible vs. Irreversible Denaturation**

- **Structure → Function Relationship:** Enzyme catalytic activity depends strictly on its native **3D folding (secondary 2° , tertiary 3° , and quaternary 4° structure)**, which is stabilized by weak noncovalent interactions:
  - **Hydrogen bonds** (backbone & polar R-groups)
  - **Ionic interactions / salt bridges** (charged acidic & basic R-groups)
  - **Hydrophobic interactions** (nonpolar core) + covalent disulfide bridges (**–S–S–**)
- **Denaturation:** Loss of native 3D protein conformation caused by extreme **temperature, pH, salinity (ion concentration), or chemical denaturants** → active site is distorted → substrate can no longer bind → **loss of catalytic activity**.
- **Reversible vs. Irreversible Denaturation:**
  - **Reversible (Renaturation):** Mild environmental shift; once optimal conditions are restored, the polypeptide spontaneously refolds into its active shape.
  - **Irreversible:** Extreme heat or pH causes permanent aggregation/precipitation (e.g., boiling an egg white albumin).

[CORE AXIOM / REMEMBER] **Denaturation = enzyme unfolds → active site loses shape → no fit for substrate!**

[COMMON AP EXAM FRQ TRAP] **Primary (1°) structure is NOT broken during denaturation!** Covalent **peptide bonds** remain intact (hydrolysis requires protease enzymes or harsh acid boiling); only weak **2°/3°/4°** folding bonds are disrupted.

§4. SUBSTRATE CONCENTRATION, SATURATION (V_{max}) & K_m

Michaelis-Menten Kinetics

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Substrate Concentration, Saturation (V_{max}) & K_m ➔ **Substrate Concentration ([S])** ➔ **Saturation & V_{max}** ➔ **Michaelis Constant (K_m — Affinity Inverse)**

- **Substrate Concentration ([S]):** At low **[S]**, empty active sites are abundant → increasing **[S]** linearly increases reaction velocity (V_0).
- **Saturation & V_{max}:** At high **[S]**, the curve plateaus at **maximum velocity (V_{max})** because **100% of enzyme active sites are saturated** with substrate. The only way to increase V_{max} is to **add more enzyme (↑ [E])**.
- **Michaelis Constant (K_m — Affinity Inverse):**
  - Defined as the **substrate concentration [S] at exactly ½ V_{max}**.
  - **Low K_m = High substrate affinity** (enzyme reaches ½ V_{max} at very low **[S]**; binds substrate tightly).
  - **High K_m = Low substrate affinity** (requires high **[S]** to reach ½ V_{max}).

[AP EXAM FRQ SCORING STRATEGY] **Plateau on a [S] vs. Rate graph = All active sites are occupied!**
Remember: Low K_m = binds substrate easily (high affinity).

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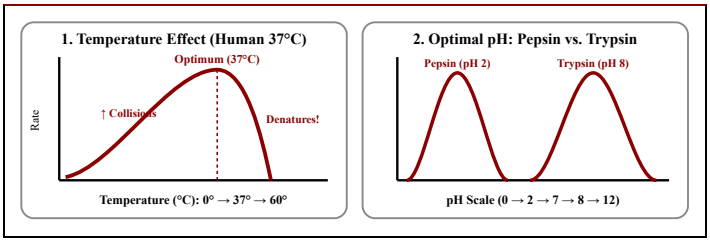
§3. TEMPERATURE & PH EFFECTS ON REACTION RATE

Asymmetric vs Bell Curves

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Temperature & pH Effects on Reaction Rate ➔
Temperature Effects (Asymmetric Curve) ➔ **pH Effects (Symmetric Bell Curves)**

- **Temperature Effects (Asymmetric Curve):**
  - **Below Optimum (↑ Temp → ↑ Rate):** Higher thermal kinetic energy increases molecular velocity and **frequency of successful enzyme-substrate collisions**.
  - **Above Optimum (Rapid Drop):** Excessive thermal vibration breaks weak H-bonds and hydrophobic interactions → **thermal denaturation** → steep plunge in activity.
- **pH Effects (Symmetric Bell Curves):**
  - Each enzyme has an **optimal pH** matching its cellular compartment.
  - Shifting **[H⁺]** protonates or deprotonates acidic (**-COO⁻**) and basic (**-NH₃⁺**) R-groups → **alters side-chain charges** → disrupts ionic bonds and active-site substrate binding.

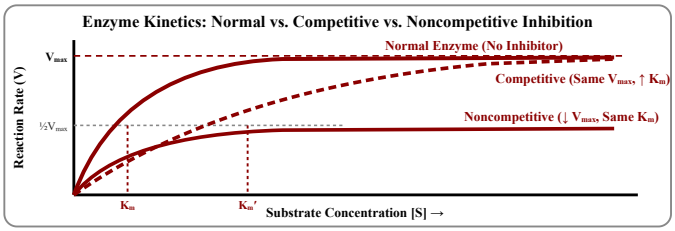


[EXAMPLE & BIOLOGICAL APPLICATION] **Compartmental pH**

Specialization: Pepsin (stomach protease) has an acidic optimum of **pH ~2**; **Trypsin** (small intestine protease) has an alkaline optimum of **pH ~8**; **Lysosomal hydrolases** work at **pH ~4.5–5** (inactive if leaked into pH 7.2 cytosol!).

§5. COMPETITIVE VS. NONCOMPETITIVE & FEEDBACK INHIBITION

Inhibitor Kinetics



INHIBITION TYPE	BINDING SITE & MECHANISM	EFFECT ON V_{max} & K_m	OVERCOME BY ↑ [S] ?
Competitive	**Active site** ("fake key blocks lock"); mimics substrate shape.	V_{max} **unchanged**; ↑ K_m (lower apparent affinity)	**YES** — flooded [S] outcompetes inhibitor.
Noncompetitive (Allosteric)	**Allosteric site**; changes enzyme 3D shape so active site cannot catalyze.	↓ V_{max} (fewer functional enzymes); K_m **unchanged**	**NO** — more substrate does NOT fix warped active site.

- **Product / Feedback Inhibition (Negative Feedback):** High concentration of a pathway's **end product** binds allosterically to the **first committed enzyme** in the pathway, shutting down synthesis to prevent waste of energy and raw materials.

[AP EXAM FRQ SCORING STRATEGY] **FRQ Golden Rule:** Always connect the full causal chain:

Environmental Change → Disrupts Weak Bonds (2°/3°) → Alters Active Site 3D Shape → ↓ ES Complex Formation → ↓ Reaction Rate!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 3.2 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 3.2: Environmental Impacts on Enzyme Function, Kinetics & Inhibition ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔
Experimental Perturbation (±2 SE_x̄) ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 5, 6 & 13 (Enzyme Catalysis, Photosynthesis Leaf Disks & Respirometry): Enzyme assay (catalase/peroxidase spectrophotometry), Floating Leaf Disk Assay (ET_{50} = time for 50% of disks to float in $NaHCO_3$), and KOH Respirometry (KOH absorbs CO_2 gas as a solid precipitate so volume decrease directly measures O_2 consumption). Negative Control: boiled enzyme/boiled seeds or glass beads (corrects for ambient temp/pressure changes). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Reaction Rate = ΔProduct / ΔTime (initial linear slope V_0). Michaelis-Menten Kinetics: Competitive inhibitors increase apparent K_m without changing V_max; Noncompetitive/allosteric inhibitors decrease V_max without changing K_m. Gibbs Free Energy: $\Delta G < 0$ (exergonic/spontaneous, coupled via ATP → ADP + P_i, $\Delta G^{\circ\prime} = -7.3$ kcal/mol) vs. $\Delta G > 0$ (endergonic). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Bioenergetic Disruption Case Study: Cyanide (CN^-) and Carbon Monoxide (CO) bind irreversibly to Complex IV (Cytochrome c oxidase) of the mitochondrial ETC, blocking electron transfer to O_2, halting proton pumping, collapsing the H^+ gradient across the inner mitochondrial membrane, and arresting ATP Synthase oxidative phosphorylation. Conversely, 2,4-Dinitrophenol (DNP) and thermogenin (UCP1) make the inner membrane permeable to H^+, uncoupling electron transport from ATP synthesis and dissipating proton-motive force as heat.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 3.2 to prove why the claim holds.

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